Euclid's Elements — Book XI
Proposition 4

Formal Reconstruction — Technical Documentation

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CGJteamLab

If a line is perpendicular at a common point to two intersecting lines of a plane, then it is perpendicular to the plane.

1 Statement

Euclid XI.4 gives the basic criterion for perpendicularity to a plane: if a line through the common point of two intersecting lines of a plane is perpendicular to each of those two lines, then it is perpendicular to the plane itself.

The formal reconstruction is implemented in

```
CGJteamLab/Proposition11_4.lean
```

on top of the spatial infrastructure in

```
CGJteamLab/Hilbert3DAxioms.lean
CGJteamLab/Hilbert3DInterface.lean
```

The Lean source is authoritative for theorem signatures, helper names, actual dependencies, and proof status.

2 Euclid XI.3 and the target notion

Book XI, Definition 3 says that a straight line is perpendicular to a plane when it makes right angles with every straight line which meets it and lies in that plane. The project encodes exactly this universal synthetic notion.

For two incidence lines:

```
def HilbertLinesPerpendicularAt
    [H : HilbertIncidence Geo]
    (l m : Geo.Line)
    (O : Geo.Point) : Prop :=
  H.OnLine O l /\
  H.OnLine O m /\
  exists A B : Geo.Point,
    Ne A O /\
    Ne B O /\
    H.OnLine A l /\
    H.OnLine B m /\
    Not (PrimCollinear Geo A O B) /\
    HilbertRightAngle Geo A O B
```

The explicit noncollinearity condition is important. It prevents a degenerate "self-perpendicular" witness and yields the reusable theorem

```
hilbert_linesPerpendicularAt_ne
```

showing that perpendicular lines are distinct.

For a line and a plane:

```
def HilbertLinePerpendicularPlaneAt
    [H : HilbertIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    (l : Geo.Line)
    (pi : S.Plane)
    (O : Geo.Point) : Prop :=
  H.OnLine O l /\
  S.OnPlane O pi /\
  forall m : Geo.Line,
    HilbertLineInPlane Geo m pi ->
    H.OnLine O m ->
    HilbertLinesPerpendicularAt Geo l m O
```

Thus the conclusion of XI.4 is not a metric abbreviation. It is literally the universal quantification required by Euclid XI.Def.3.

3 Classical configuration

Let m, n be two distinct straight lines of a plane π, meeting at O, and let l be a line through O perpendicular to both.

The desired conclusion is that l is perpendicular to every line $q \subset \pi$ through O.

4 Euclid's classical proof

Euclid makes the argument constructive.

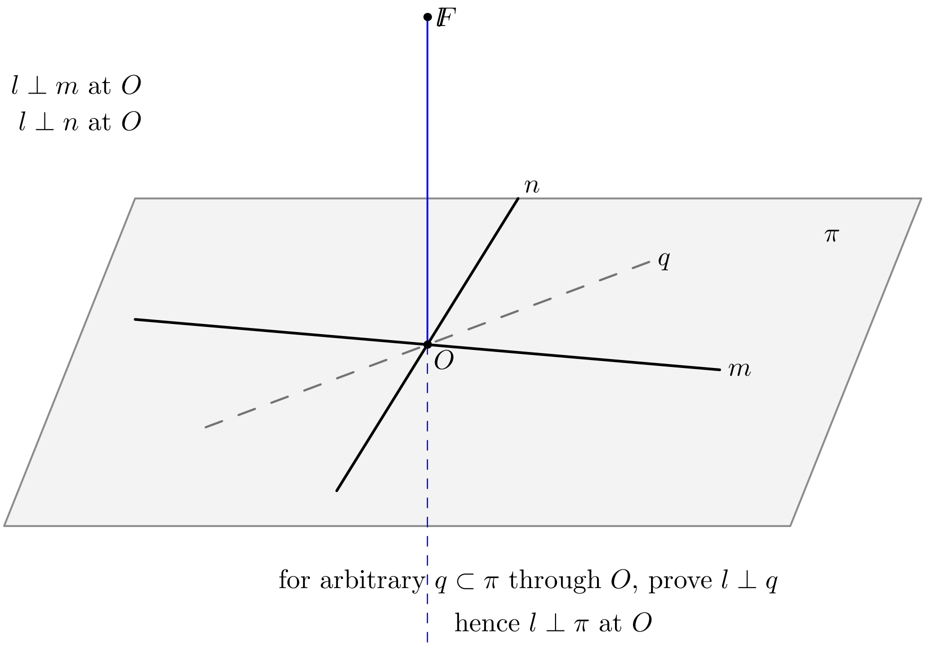


Figure 1: Classical configuration for Euclid XI.4. The ambient line l passes through O and is perpendicular at O to two distinct lines m, n lying in the plane π. The dashed line q represents an arbitrary line of π through O, which is the universal target of the conclusion.

Take equal segments on the two given plane lines:

$$AE = EB = CE = ED,$$

where E is the common point of the two given lines. Draw an arbitrary straight line GEH through E in the plane, choose an arbitrary point F on the elevated perpendicular, and join the auxiliary segments.

The classical dependency chain is:

1. Vertical angles at E are equal (I.15). Together with the four equal radial segments, I.4 gives

   $$AD = CB$$

   and a corresponding angle equality.
2. Applying I.26 to the planar transversal gives

   $$GE = EH, \qquad AG = BH.$$

3. Since FE is perpendicular to both original plane lines, I.4 applied to two right triangles gives

   $$FA = FB, \qquad FC = FD.$$

4. I.8 compares two spatial triangles and supplies the angle relation needed to transport the planar data to the arbitrary transversal.
5. A further I.4 gives

   $$FG = FH.$$

6. Finally, from
$$GE = EH, \qquad FE = FE, \qquad FG = FH,$$
I.8 yields
$$\angle GEF = \angle HEF.$$
Since $G - E - H$, the adjacent equal angles are right angles.

7. The transversal GEH was arbitrary. Hence FE makes a right angle with every line through E in the plane. By XI.Def.3, $FE \perp \pi$.

This is a genuinely three-dimensional use of Book I congruence theory: several compared triangles are not assumed to lie in one common plane.

5 Formal theorem

The production theorem is:

```
theorem euclid_proposition_11_4
    [H : HilbertIncidence Geo]
    [HilbertPlaneIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HSI : HilbertSpaceIncidence Geo]
    [_HSO : HilbertSpaceOrder
      (Geo := Geo) (H := H) (S := S)]
    [HSC : HilbertSpaceCongruence
      (Geo := Geo) (H := H) (S := S)]
    (pi : S.Plane)
    (m n : PlaneLine Geo pi)
    (l : Geo.Line)
    (O : PlanePoint Geo pi)
    (hmn : Ne m n)
    (hPerpM :
      HilbertLinesPerpendicularAt Geo l m.1 O.1)
    (hPerpN :
      HilbertLinesPerpendicularAt Geo l n.1 O.1) :
    HilbertLinePerpendicularPlaneAt Geo l pi O.1
```

The theorem does not assume separately that $l \neq m$ or $l \neq n$. Those facts follow from the strengthened nondegenerate line–line perpendicularity predicate.

6 Spatial architecture

6.1 No ambient planar type-class instance

The ambient geometry is three-dimensional. The proof deliberately does *not* install a global instance

```
HilbertCongruence Geo
```

or

```
HilbertOrder Geo
```

on the whole space. Planar reasoning is performed only in explicit plane slices

$$\mathrm{PlaneGeo}(\pi).$$

The corresponding points and lines are subtypes:

```
PlanePoint Geo pi
PlaneLine  Geo pi
```

and the interface provides bridges between induced planar facts and ambient facts, for example

```
planeGeo_congruent
planeGeo_angleCongruent_iff_ambient
planeGeo_sameRay_iff_ambient
planeGeo_rightAngle_iff_ambient
planeGeo_linesPerpendicularAt_iff_ambient
```

This separation is one of the central architectural features of the proof.

6.2 Spatial congruence

Euclid freely applies I.4 and I.8 to triangles in different planes. The project reconstructs this through the spatial congruence layer rather than by coordinates or by an ambient Euclidean metric.

The general spatial infrastructure includes:

```
hilbert_space_congruent_reflexive
hilbert_space_congruent_symmetry
hilbert_space_sas_remaining_angles
hilbert_space_sas_third_side_and_angle
hilbert_space_sss_angleA_in_plane
```

The target triangle may be placed in its own explicit plane and compared with an ambient source triangle. This is the formal counterpart of Euclid's unqualified use of plane triangle congruence inside solid geometry.

7 The balanced cross

The first construction is the formal analogue of Euclid's

$$AE = EB = CE = ED.$$

The helper

```
planeGeo_opposite_points_on_line_congruent_to
```

constructs opposite equal points on one line. Applying it twice yields

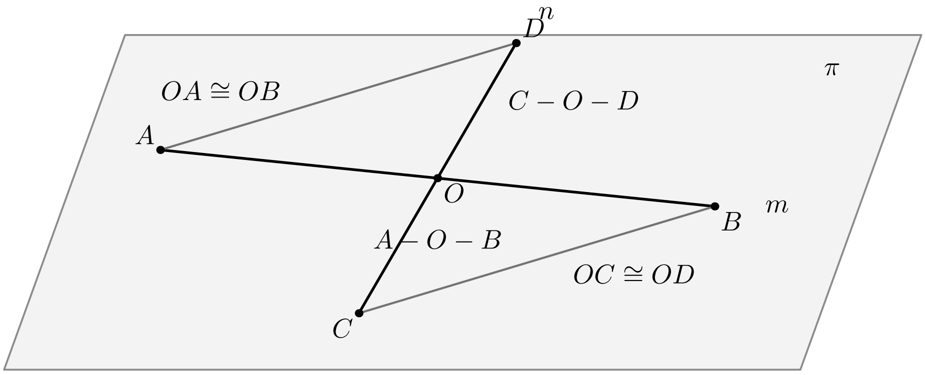
```
hilbert_XI4_balanced_cross_exists
```

and points A, B, C, D satisfying

$$A - O - B, \qquad C - O - D$$

and

$$OA \cong OR, \quad OB \cong OR, \quad OC \cong OR, \quad OD \cong OR.$$



all objects in this figure lie in π

Figure 2: Balanced cross in π. Points A, B lie on m, points C, D lie on n, with $A - O - B$, $C - O - D$, $OA \cong OB$, and $OC \cong OD$. The connector segments AD and BC are the first pair used by Euclid's congruence argument.

The first triangle comparison is isolated by

```
planeGeo_XI4_cross_noncollinear
hilbert_XI4_balanced_cross_first_SAS
```

and proves the ambient consequences

$$AD \cong BC, \qquad \angle OAD \cong \angle OBC.$$

This is the formal Hilbert version of Euclid's first use of I.4.

8 Fixed-transversal prototype

Before the proof was generalized to every transversal, a fixed connector configuration was developed. Its role is mathematically transparent: given

$$A - G - D, \qquad B - J - C, \qquad G - O - J,$$

the balanced cross implies

$$OG \cong OJ, \qquad AG \cong BJ.$$

The relevant planar ingredients are

```
planeGeo_XI4_transversal_noncollinear
hilbert_XI4_transversal_ASA
```

corresponding to Euclid's I.26 step.

The spatial point $F \in l$, together with the two original perpendicularities, then gives

$$AF \cong BF, \qquad CF \cong DF.$$

The proof packages this in

```
hilbert_XI4_spatial_perpendicular_bisector_equidistant
hilbert_XI4_spatial_opposite_pairs_equidistant
```

The spatial SSS/SAS chain is then:

```
hilbert_XI4_spatial_SSS_angle
hilbert_XI4_space_angle_sameRay_first_congruent
hilbert_XI4_spatial_SSS_angle_normalized
hilbert_XI4_spatial_SAS_transversal_side
hilbert_XI4_spatial_transversal_F_equidistant
hilbert_XI4_second_spatial_SSS_angle
hilbert_XI4_right_angle_of_adjacent_equal
```

Its mathematical content is

$$\begin{gathered} AD \cong BC, \quad AF \cong BF, \quad CF \cong DF \\ \Downarrow \\ \angle DAF \cong \angle CBF \\ \Downarrow \\ A - G - D,\ B - J - C \\ \Downarrow \\ \angle GAF \cong \angle JBF \\ \Downarrow \\ AG \cong BJ,\ AF \cong BF \\ \Downarrow \\ GF \cong JF \\ \Downarrow \\ OG \cong OJ,\ OF \cong OF,\ GF \cong JF \\ \Downarrow \\ \angle GOF \cong \angle JOF \\ \Downarrow \\ G - O - J \\ \Downarrow \\ \mathrm{RightAngle}(G, O, F). \end{gathered}$$

The last implication is exactly Euclid's definition of a right angle as two adjacent equal angles formed by a straight line.

9 The arbitrary transversal: the real formal difficulty

The public conclusion quantifies over *every* line of π through O. A conventional drawing suggests that such a transversal meets the two connector segments AD and BC, but that is false in general. This was the key positional issue in the formal proof.

The correct argument uses two possible connector families:

$$AD/BC \qquad \text{or} \qquad BD/AC.$$

For a transversal $q \neq m, n$, apply Pasch to the triangle ABD. The line q enters through AB at O, hence it exits either through AD or through BD.

This is isolated as

```
hilbert_XI4_transversal_pasch_first_exit
```

Next,

```
hilbert_XI4_opposite_point_on_transversal
```

constructs the point J opposite G on q with

$$G - O - J, \qquad OG \cong OJ.$$

Two neutral landing lemmas then prove:

```
hilbert_XI4_first_exit_opposite_lands_on_BC
hilbert_XI4_second_exit_opposite_lands_on_AC
```

so that the opposite point lands on the correct paired connector. The combined package is

```
hilbert_XI4_arbitrary_transversal_connector_package
```

and returns

$$\begin{gathered} G, J \in q, \quad G - O - J, \quad OG \cong OJ, \\ \text{and either} \\ A - G - D,\ B - J - C, \\ \text{or} \\ B - G - D,\ A - J - C. \end{gathered}$$

This entire stage uses incidence, order, Pasch, and congruence. It does not use the Euclidean parallel axiom.

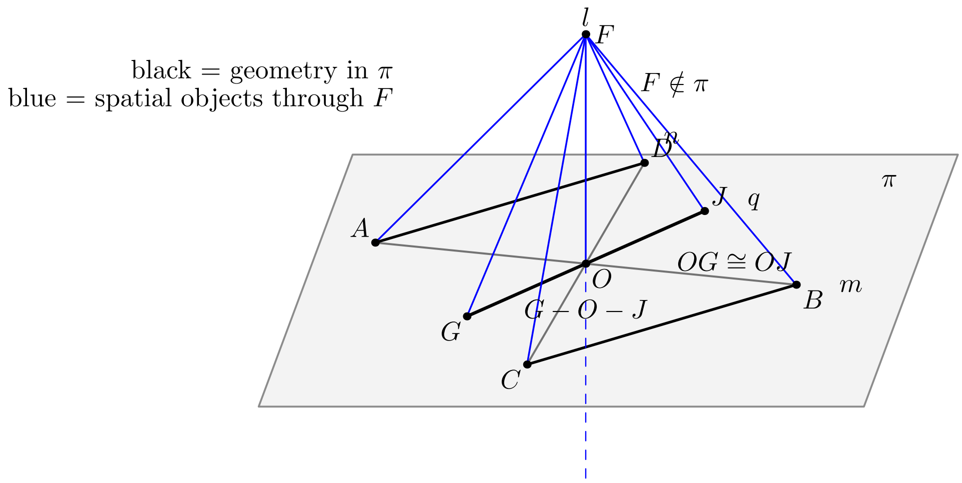


Figure 3: Arbitrary-transversal stage. For $q \subset \pi$ through O, Pasch selects one of the two connector families. The opposite point J is constructed with $G - O - J$ and $OG \cong OJ$. The spatial SSS/SAS chain then proves that $\angle GOF$ is right.

10 Right angle on every transversal

With the connector package available, both branches reduce to the same spatial SSS/SAS scheme. The theorem

```
hilbert_XI4_arbitrary_transversal_right_angle
```

states, in substance:

$$q \subset \pi, \quad O \in q, \quad q \neq m, n,$$
$$F \notin \pi, \quad AF \cong BF, \quad CF \cong DF$$

imply the existence of $G \in q$, $G \neq O$, such that

$$\text{RightAngle}(G, O, F).$$

The two exceptional cases $q = m$ and $q = n$ are discharged directly from the hypotheses of XI.4.

11 Why the point F is outside the plane

The core proof chooses an arbitrary nonvertex point F on l. To use spatial noncollinearity cleanly, it proves

$$F \notin \pi.$$

This is the theorem

```
hilbert_XI4_perpendicular_point_off_plane
```

The proof is neutral and conceptually important.

Assume $F \in \pi$. Then the line l, determined by O and F, is itself a line of π. Transfer the two ambient perpendicularities to `PlaneGeo pi`. Choose nonvertex points $A \in m$ and $C \in n$. Then both

$$\angle FOA \quad \text{and} \quad \angle FOC$$

are right.

The helper

```
hilbert_XI4_two_right_angles_same_first_arm_collinear
```

uses Hilbert III.4 uniqueness to prove that A, O, C are collinear. Line uniqueness then forces $m = n$, contradicting the hypothesis.

No parallel postulate is used.

12 Closing the universal line–plane statement

The core theorem

```
hilbert_XI4_line_perpendicular_plane_core
```

does the following:

1. choose $F \neq O$ on l;
2. construct the balanced cross A, B, C, D in π;
3. prove $F \notin \pi$;
4. prove $AF \cong BF$ and $CF \cong DF$;
5. let q be an arbitrary line of π through O;
6. if $q = m$ or $q = n$, use the hypotheses;
7. otherwise invoke the arbitrary-transversal theorem and obtain a point $G \in q$ such that $\angle GOF$ is right;
8. construct the witness required by `HilbertLinesPerpendicularAt Geo l q O`.

There is one subtle final representation step. The right angle obtained from the transversal theorem is oriented as

$$\angle GOF.$$

The line-line perpendicularity witness is more convenient with the order

$$\angle FOG.$$

The proof does not use an ambient planar congruence instance to swap the arms. Since l and q intersect at O, they determine an explicit plane ρ. Inside `PlaneGeo rho` the right angle is swapped using the ordinary planar congruence API, and the result is transported back to the ambient geometry.

This preserves the 3D architecture: all planar congruence reasoning occurs inside an explicit plane slice.

13 Complete formal DAG

The proof architecture can be summarized as follows.

```
two distinct plane lines m,n through O
+ l perpendicular to m and n at O
-> balanced cross A-O-B, C-O-D
-> AD  = BC and angle OAD  = angle OBC
-> choose F != O on l
-> F notin pi
-> AF  = BF, CF  = DF
-> arbitrary q in pi through O
-> if q=m or q=n:  hypothesis
-> otherwise Pasch in triangle ABD
-> exit through AD or BD
-> opposite J on q, G-O-J, OG  = OJ
-> landing on BC or AC
-> ASA: AG  = BJ (or BG  = AJ)
-> spatial SSS/SAS: GF  = JF
-> spatial SSS: angle GOF  = angle JOF
-> G-O-J -> RightAngle GOF
-> l perpendicular q at O
-> forall q in pi through O, l perpendicular q
-> HilbertLinePerpendicularPlaneAt Geo l pi O.
```

14 Classical dependencies versus Lean dependencies

Role	Euclid	Lean reconstruction
Four equal radial segments	cut off AE, EB, CE, ED equal	opposite-point construction on each of the two lines inside `PlaneGeo pi`
Vertical angles	I.15	developed vertical-angle and angle-congruence infrastructure
First triangle congruence	I.4	balanced-cross SAS helper
Transversal equalities	I.26	Pasch, transversal ASA, and the two landing branches
Distances from F	I.4 on right triangles	spatial perpendicular-bisector/equidistance helper
Cross-plane triangle comparison	I.8 and I.4	general spatial SSS/SAS API through explicit carrier planes
Arbitrary line in the plane	“drawn at random”	universal q; cases $q = m$, $q = n$, or the Pasch branch
Line perpendicular to plane	XI.Def.3	the universal line–plane perpendicularity predicate

The Lean DAG is longer because it must prove all incidence, order, noncollinearity, carrier-plane, and orientation facts that the classical diagram leaves implicit. The geometric argument itself remains Euclidean XI.4.

15 Axiomatic status

The public theorem assumes the spatial Hilbert hierarchy

```
HilbertIncidence Geo
HilbertPlaneIncidence Geo
HilbertSpacePrimitive Geo
HilbertSpaceIncidence Geo
HilbertSpaceOrder
HilbertSpaceCongruence
```

No `HilbertEuclideanPlane` instance occurs in the theorem signature. In particular, the proof of XI.4 does not use Hilbert Group IV, the Euclidean parallel axiom.

No continuity or Archimedean axiom is introduced by XI.4.

The final Lean audit is:

```
'Geometry.euclid_proposition_11_4' depends on axioms:
[propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

The project deliberately retains the inherited `bookZero_nullSegment1` dependency at this stage. XI.4 introduces no new proposition-specific axiom.

16 Formalization notes

16.1 The proof is synthetic

There are no coordinates, vectors, scalar products, normal vectors, trigonometric functions, angle measures, or real-valued lengths.

All claims are expressed through:

- incidence of points, lines, and planes;
- strict betweenness;
- same-ray and side-of-line relations;
- segment and angle congruence;
- Hilbert right angles;
- Pasch;
- planar and spatial SAS/SSS/ASA arguments.

16.2 The point F is not a normal-vector surrogate

The point F is merely a nonvertex point chosen on the candidate line l. It is used to form ordinary synthetic triangles such as AOF, BOF, GOF, and JOF. The proof never assigns coordinates to F and never interprets OF as a vector.

16.3 Why `PlaneGeo` is essential

Some arguments are intrinsically planar: Pasch, same-side uniqueness of an angle copy, same-ray normalization, and ordinary Hilbert angle construction. Other arguments compare triangles in different planes. The proof therefore alternates explicitly between:

$$\text{ambient 3D geometry} \quad \longleftrightarrow \quad \text{PlaneGeo}(\pi) \quad \text{or} \quad \text{PlaneGeo}(\rho).$$

This is not implementation noise. It records exactly where Euclid is using plane geometry inside a solid-geometric proof.

16.4 Hidden positional data

The formal proof makes explicit all of the following facts:

- the two given plane lines are genuinely distinct;
- perpendicularity witnesses are nondegenerate;
- the four balanced points lie on the intended carriers;
- $A - O - B$ and $C - O - D$ are strict orders;
- the auxiliary point F is outside π;
- a general transversal need not meet the fixed pair AD, BC;
- Pasch selects one of two connector families;
- the opposite point on the transversal lies on the paired connector;
- every spatial triangle used by SSS or SAS is noncollinear;
- the final arm swap is performed in the explicit plane through l and the arbitrary transversal.

These facts are precisely the information supplied visually, but not logically, by a conventional diagram.

16.5 Why the two connector families matter

A tempting but incorrect formalization is to assume that every line through O meets both segments AD and BC. The diagram can make this appear plausible, but it is not true.

The final proof instead analyzes the first exit of the transversal from triangle ABD. This produces exactly the dichotomy required by the geometry:

$$(AD, BC) \quad \text{or} \quad (BD, AC).$$

This is one of the most significant proof-design corrections in the XI.4 development.

17 Reusable infrastructure produced by XI.4

The proposition required one general correction to the 3D interface: line-line perpendicularity now carries an explicit noncollinearity witness, so that distinctness of the carriers is a theorem rather than an external hypothesis.

The general 3D interface also exposes the bridges

```
HilbertLinesPerpendicularAt
hilbert_linesPerpendicularAt_ne
planeGeo_linesPerpendicularAt_iff_ambient
HilbertLinePerpendicularPlaneAt
HilbertLinePerpendicularPlaneAt.perpendicular_to_line
HilbertLinePerpendicularPlaneAt.incidence
```

The remaining declarations beginning with `hilbert_XI4_` are kept in `Proposition11_4.lean` because they package the concrete Euclidean XI.4 configuration.

18 Dependency summary

$$\begin{gathered}
\text{Hilbert spatial incidence + order + congruence}\\
\Downarrow\\
\text{PlaneGeo slices and ambient/plane bridges}\\
\Downarrow\\
\text{nondegenerate line-line perpendicularity}\\
\Downarrow\\
\text{balanced cross } A - O - B,\ C - O - D\\
\Downarrow\\
AD \cong BC,\ \angle OAD \cong \angle OBC\\
\Downarrow\\
F \in l,\ F \neq O,\ F \notin \pi\\
\Downarrow\\
AF \cong BF,\ CF \cong DF\\
\Downarrow\\
\text{Pasch for arbitrary } q \subset \pi \text{ through } O\\
\Downarrow\\
\text{two connector branches + opposite point } J\\
\Downarrow\\
GF \cong JF, \quad \angle GOF \cong \angle JOF\\
\Downarrow\\
\mathrm{RightAngle}(G, O, F)\\
\Downarrow\\
l \perp q \text{ for every } q \subset \pi \text{ through } O\\
\Downarrow\\
\texttt{euclid_proposition_11_4}.
\end{gathered}$$

19 Status

Production theorem	`Geometry.euclid_proposition_11_4`
Production file	`CGJteamLab/Proposition11_4.lean`
General 3D interface	`CGJteamLab/Hilbert3DInterface.lean`
Coordinates/vectors	no
Euclidean parallel axiom (Group IV)	no
Continuity	no new use
New XI.4-specific axiom	no
Full project build	clean
Axiom audit	completed

20 Conclusion

Euclid XI.4 is the first major test of the project's spatial Hilbert architecture. The proposition begins with only two line-line perpendicularities at a common point and must end with a universal statement about every line of a plane through that point.

The formal proof shows that this transition can be reconstructed synthetically. The essential tools are not coordinates or vector orthogonality, but the same ingredients visible in Euclid's proof: equal-segment constructions, vertical angles, triangle congruence, an arbitrary transversal, and the definition of a perpendicular to a plane.

Lean exposes two pieces of structure that the classical diagram conceals. First, the arbitrary transversal requires a genuine Pasch argument with two connector families. Second, the repeated uses of Book I congruence in solid geometry require an explicit mechanism for comparing triangles carried by different planes.

Once these points are made explicit, the final statement is exactly Euclid XI.Def.3: the candidate line is perpendicular to every line of the given plane through the common point. In that sense the proof is not a modern replacement of XI.4 but a detailed synthetic reconstruction of its logical content.